Which one of the following will not lead to a disorder associated with an imprinted gene?

A. mutations affecting the non-imprinted copy

The non-imprinted copy of an imprinted gene is the only one that is expressed; mutating it will lead to problems

B. epimutations (defects in histone tail modification or DNA methylation that affect gene expression).

Obviously, mistakes in DNA methylation could lead to what should be an imprinted allele being expressed (or what should be a non-imprinted allele being silenced), which could be problematic

C. uniparental disomy (Robertsonian carrier, non-disjunction, somatic cell recombination).

if both copies of a chromosomes come from e.g. mom, the offspring would lack a chromosome with the paternal imprinting pattern. Any maternally imprinted genes on that chromosome would not be expressed in the offspring.

D. all three mechanisms could affect imprinting

CORRECT ANSWER

Most mitochondrial proteins are encoded by nuclear genes. Which of the following statements about the inheritance of these genes is true?

A. they will be inherited in a non-Mendelian manner

True of mtDNA genes, but NOT of nuclear genes, even if they encode mitochondrial proteins

B. only the mother's genotype will matter for the phenotype of the offspring

True for mtDNA genes, but NOT for nuclear genes

C. the mother's alleles will act in a dominant manner

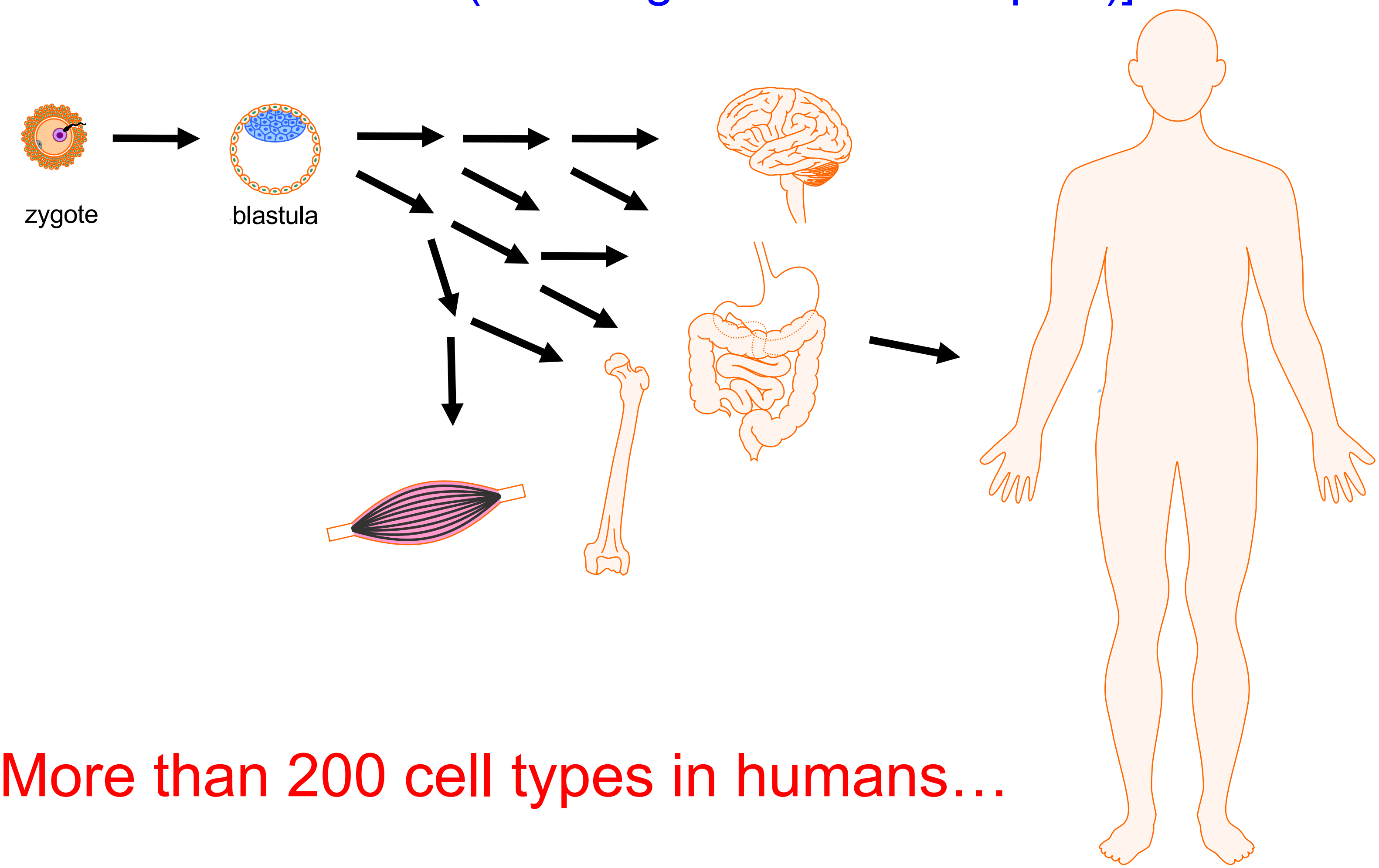
true in the case of mtDNA mutations, but not necessarily for mutations in nuclear genes

D. a wild-type paternal allele will most likely complement a defective maternal allele

TRUE (since inheritance of nuclear mitochondrial genes obeys Mendel's laws)

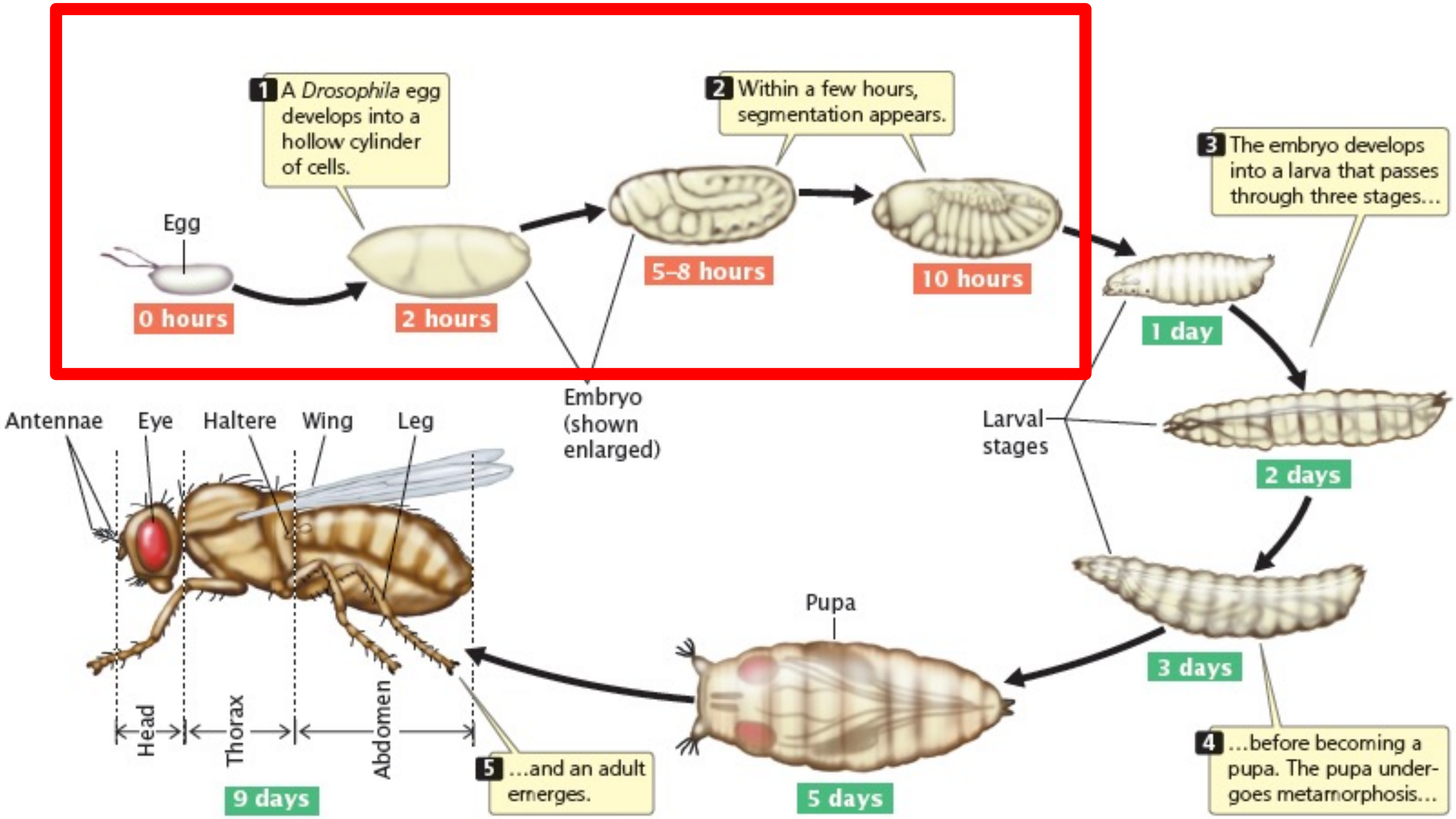
Gene regulation in embryonic development

Readings: Griffiths Chapter 13, pages 427-448 [sections Intro - 13.4 (not integration of Hox inputs)]



More than 200 cell types in humans…

Drosophila – A simpler model for deciphering developmental mechanisms

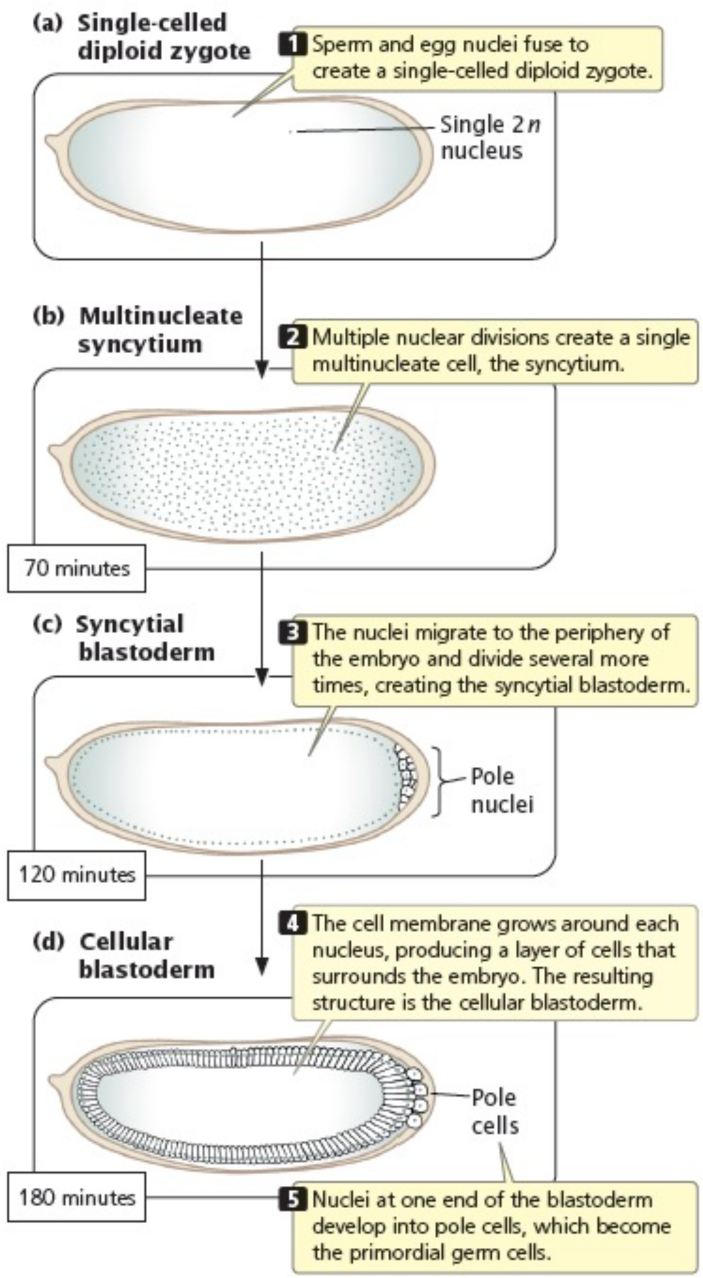


22.4 The fruit fly *Drosophila melanogaster* passes through three larval stages and a pupa before developing into an adult fly. The three major body parts of the adult are head, thorax, and abdomen.

Embryonic development in *Drosophila* (24h)

Nuclei divide many times within a single common cytoplasm

"Cellularization": individual nuclei are enclosed in plasma membrane to form cells



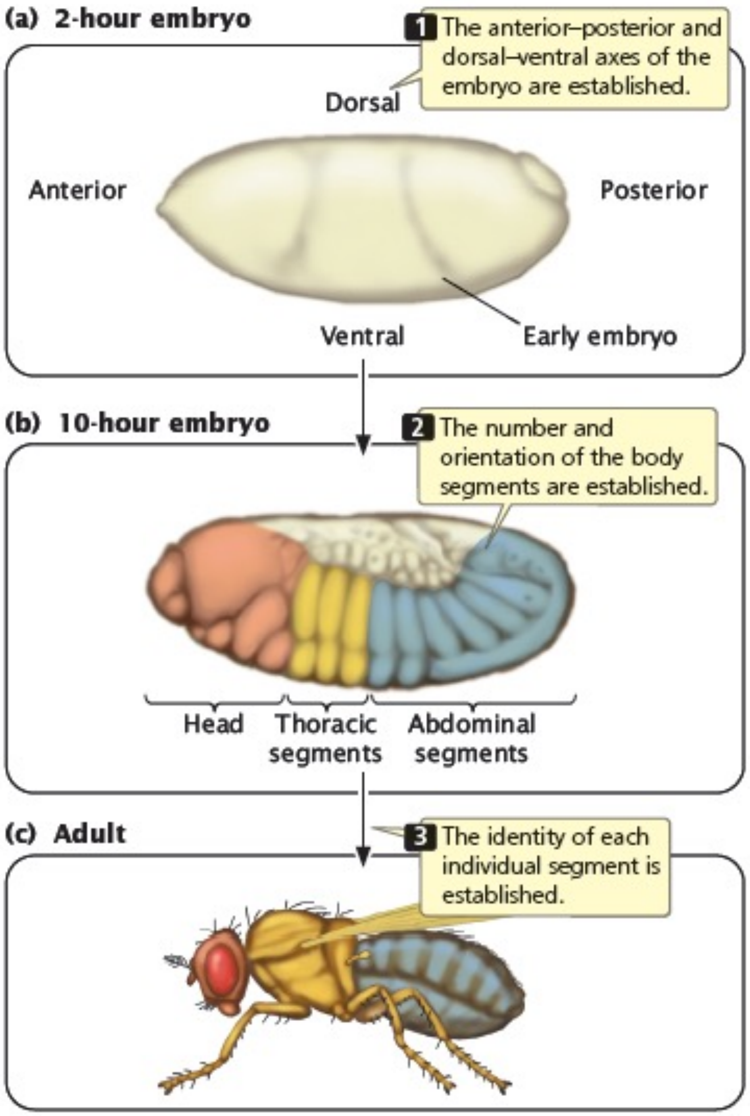
22.5 Early development of a *Drosophila* embryo.

Pierce

No transcription; development controlled entirely by mRNAs provided by mother through oocyte cytoplasm

Transcription of embryo's own genes begins (= zygotic transcription)

Embryonic development in *Drosophila*



22.6 In an early *Drosophila* embryo, the major body axes are established, the number and orientation of the body segments are determined, and the identity of each individual segment is established. Different sets of genes control each of these three stages.

after just 2 hours, the embryos has already defined head, tail, back, and front regions

body plan gets subdivided into “segments”

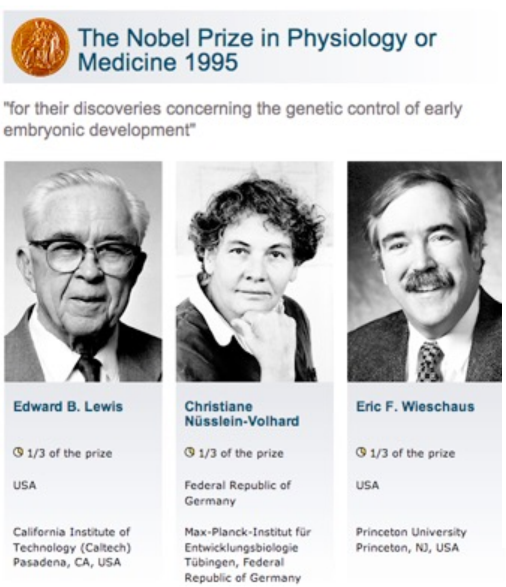
each segment will go on to form specific structures

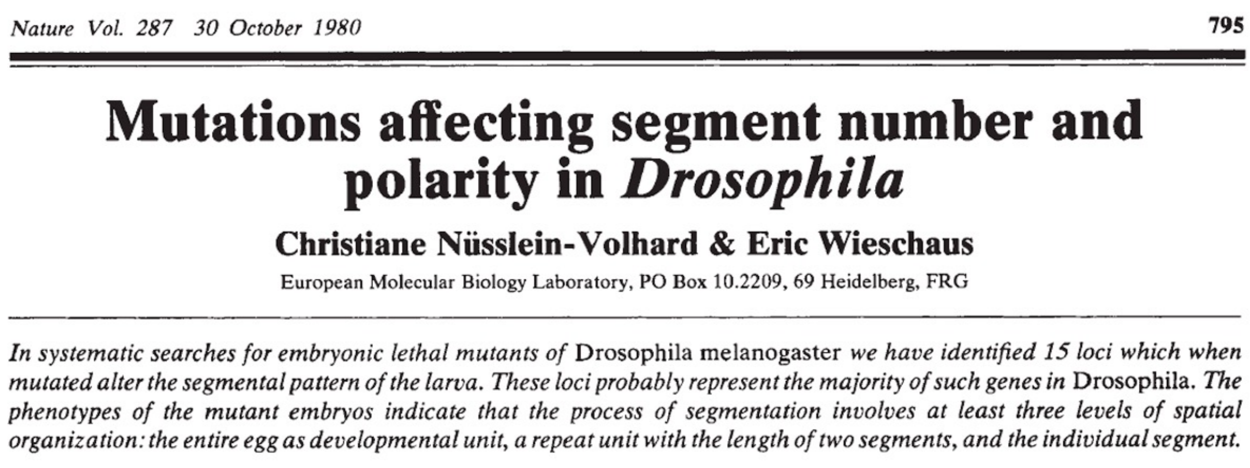
Pierce

The "Heidelberg Screen" for mutations disrupting early embryonic patterning

- Mid-1970s - Eric Wieschaus and Christine Nusslein-Volhard carried out a forward genetic screen to identify genes required for organizing the Drosophila embryo

- First systematic mutagenesis screen in a multicellular organism





Nature Vol. 287 30 October 1980 795

Mutations affecting segment number and polarity in *Drosophila*

Christiane Nüsslein-Volhard & Eric Wieschaus

European Molecular Biology Laboratory, PO Box 10.2209, 69 Heidelberg, FRG

In systematic searches for embryonic lethal mutants of Drosophila melanogaster *we have identified 15 loci which when mutated alter the segmental pattern of the larva. These loci probably represent the majority of such genes in* Drosophila. *The phenotypes of the mutant embryos indicate that the process of segmentation involves at least three levels of spatial organization: the entire egg as developmental unit, a repeat unit with the length of two segments, and the individual segment.*

Two screens: Maternal vs zygotic genes

- Screen 1: What mutations **in the mother** prevent her offspring from completing embryonic development?
- (put another way – what gene products (mRNAs) does a fly mother need to provide to her offspring through the oocyte cytoplasm in order for them to complete early embryonic development, before they start transcribing their own genes?
- **cytoplasmic inheritance (= maternal inheritance)**
  - **Maternal-effect genes:**
    "offspring's phenotype determined by ***MOTHER'S*** *genotype"*

(a bit like what we saw with mtDNA gene mutations)

MATERNALLY REQUIRED GENES

Parents		Offspring	
$m/+$ ♂ × $m/+$ ♀	⟶	$m/m, m/+, +/+$	all normal
m/m ♂ × $m/+$ ♀	⟶	$m/m, m/+$	all normal
$+/+, m/+,$ or m/m ♂ × **m/m♀**	⟶	$m/+, m/m$	**all mutant phenotype**

Fig 13-12

Two screens: Maternal vs zygotic genes

- Screen 2: Which of <u>the embryo's own genes</u> are needed for it to develop normally?

  - **zygotic genes** – *phenotype determined by genotype of the embryo*
  - inheritance shows "normal" Mendelian pattern

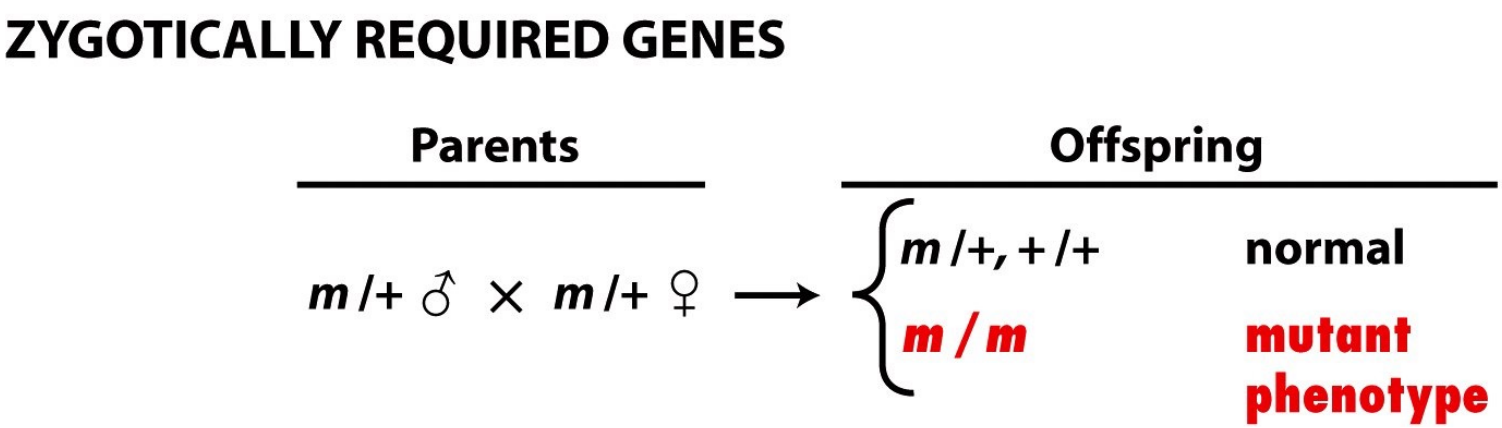
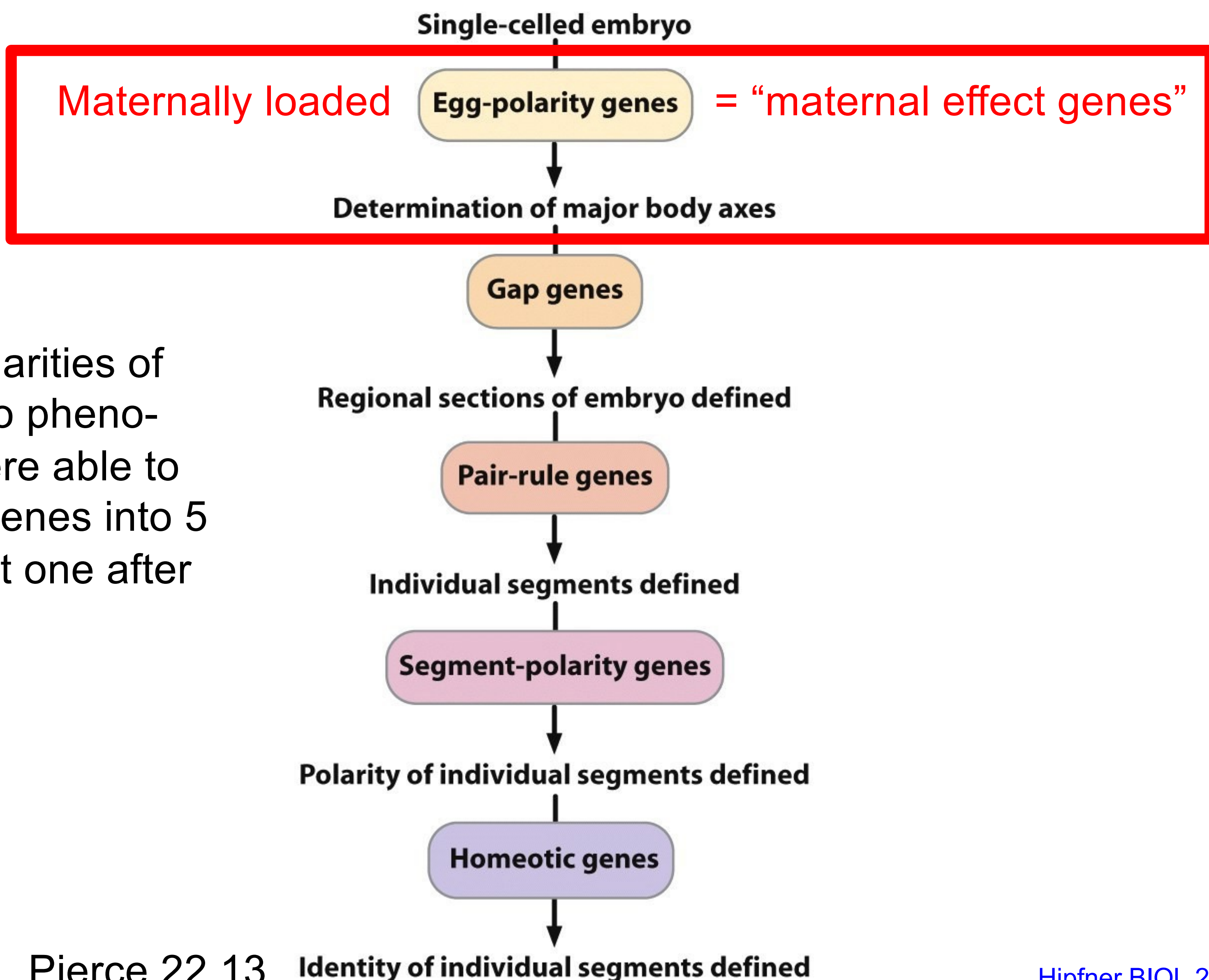


Fig 13-12

Cascade of transcription factor expression regulates early *Drosophila* development



based on similarities of mutant embryo pheno-types, they were able to organize the genes into 5 groups that act one after the other

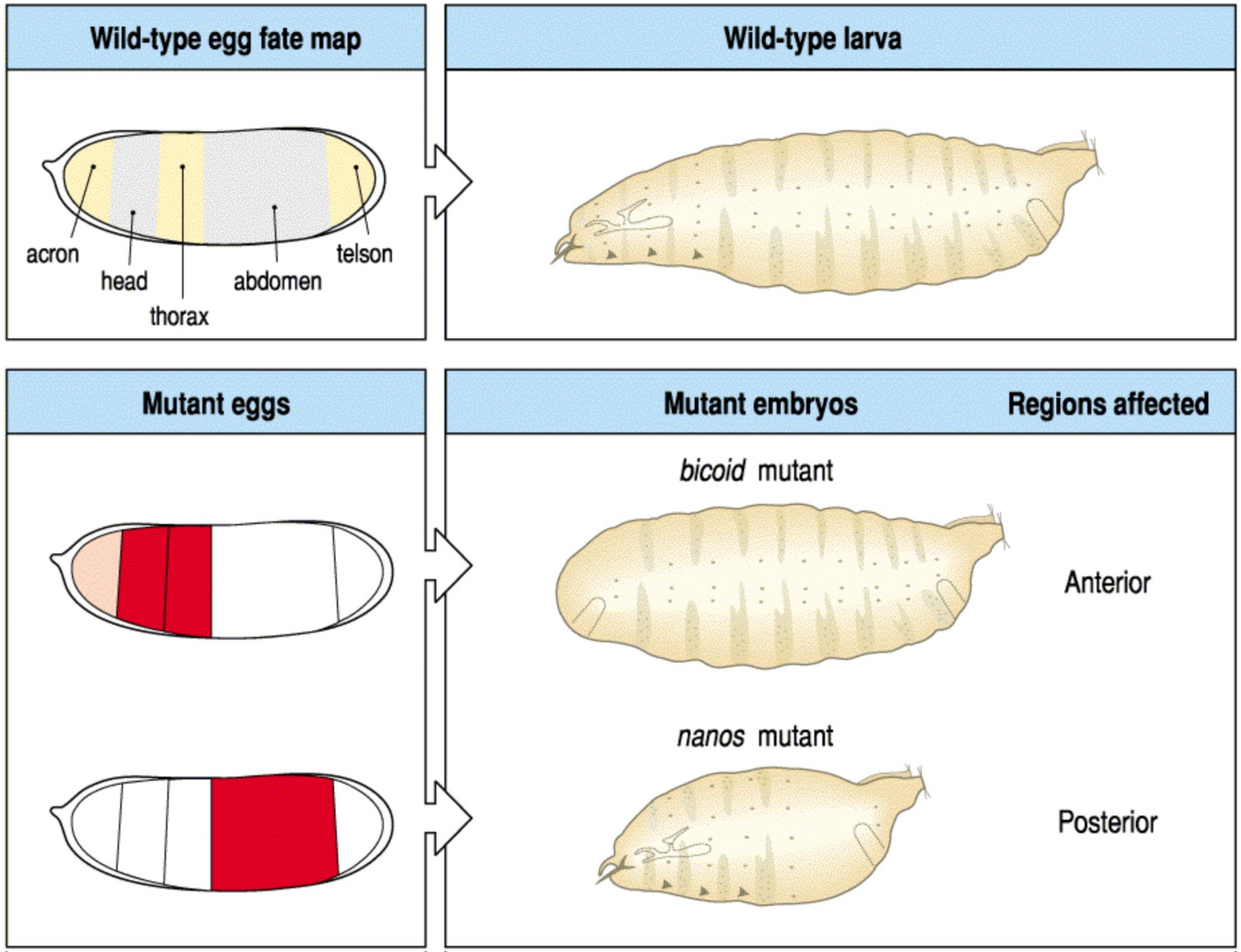
Pierce 22.13

Maternal-effect genes establish A-P axis

Two maternal-effect genes specify the anterior-posterior axis:

Anterior (A): bicoid
mutants lack anterior structures (head, thoracic segments)

Posterior (P): nanos
mutants lack posterior (abdominal) segments



Wolpert, Fig 5.3

Anterior-posterior maternal-effect genes

- *bicoid* (*bcd*) mutants lack anterior segments
  - role in establishment of basic positional information along anterior-posterior axis
- encodes a transcription factor
  - protein forms A [high] → P [low] concentration gradient

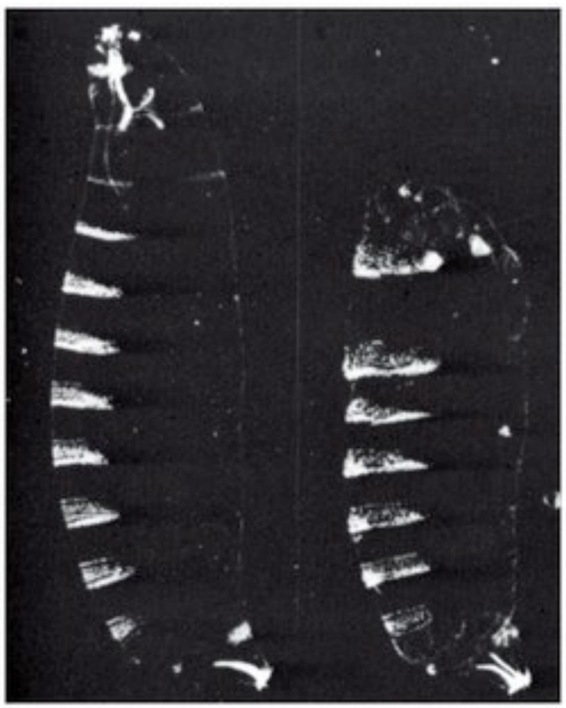


Fig 13-13

Bicoid proteins

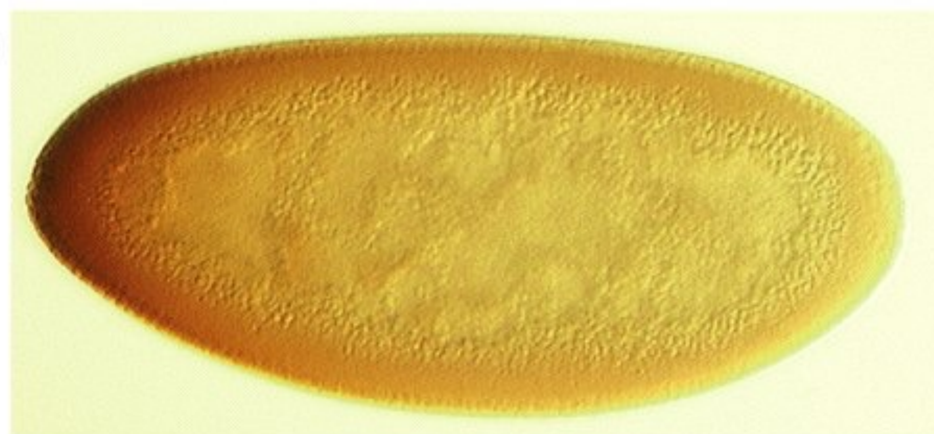
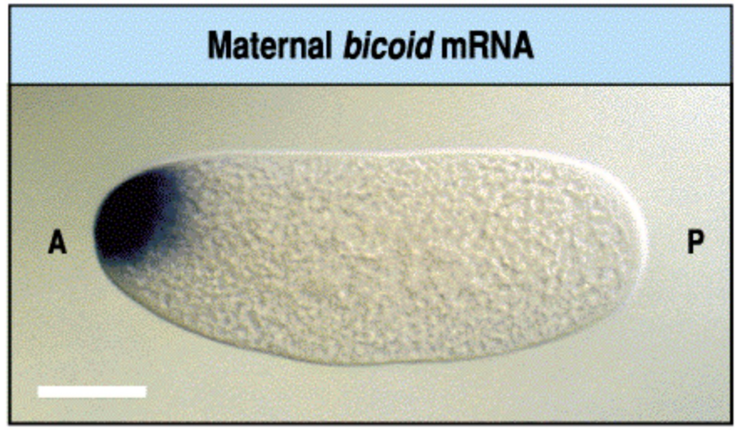


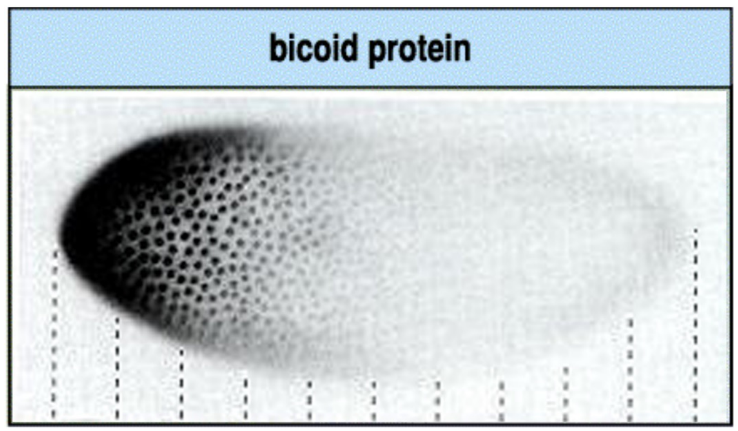
Fig 13-15a

The anterior Bicoid protein gradient

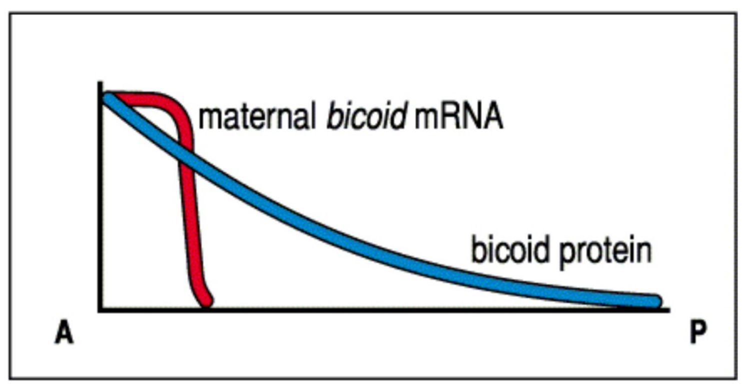
bicoid mRNA (**from mother**) localizes to anterior of embryo



Bicoid protein translated in anterior from maternal mRNA



Bicoid proteins diffuse toward the posterior region, making a **concentration gradient** – <u>providing information about distance from anterior</u>

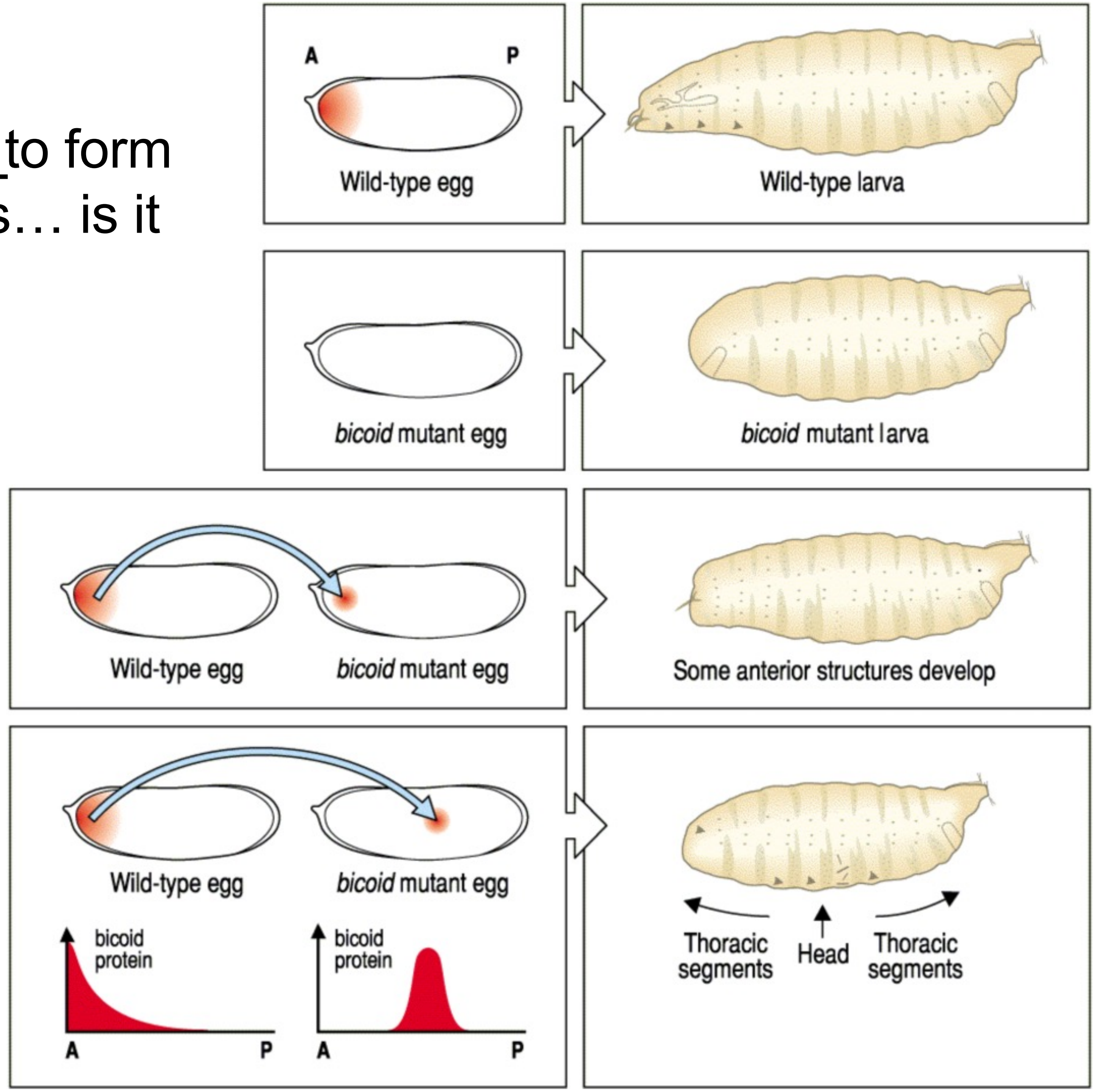


Modified from Wolpert Gilbert

Bicoid directs formation of anterior structures

Bicoid is <u>required</u> to form anterior structures… is it **sufficient**?

YES – Injected *bicoid* mRNA directs formation of anterior structures

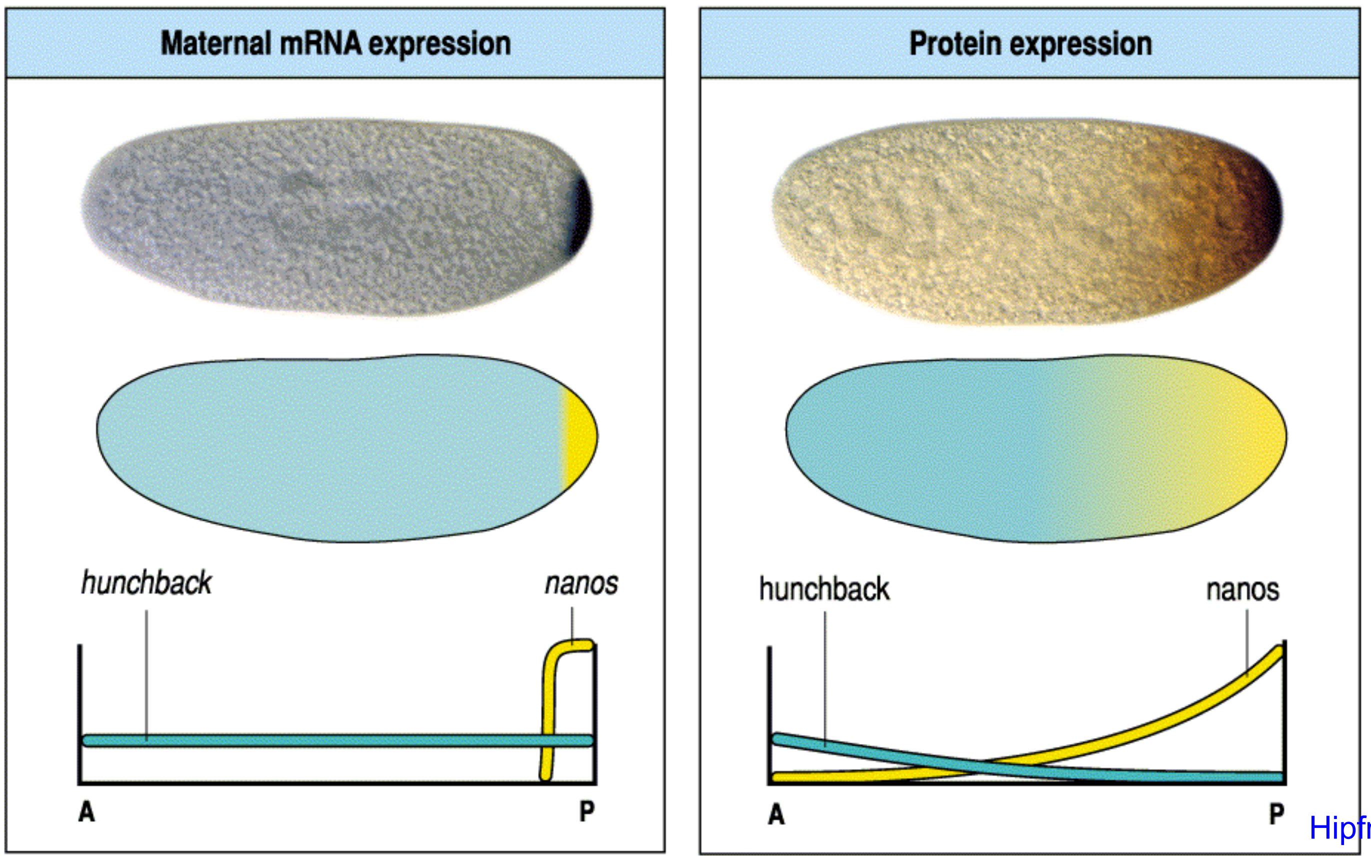


Nanos and maternal Hunchback pattern the posterior

• maternal *nanos* mRNA is localized to the posterior end of the embryo, translated protein forms concentration gradient from A [low] to P [high]

• Nanos **inhibits translation** of uniformly-distributed maternal *hunchback* mRNA

• creates an A [high] to P [low] concentration gradient of the transcription factor Hunchback



What would be the appearance of offspring of the cross between a *nanos-/-* male and a *bicoid+/-* female?

A. 25% of offspring have no head (anterior)

B. 100% of offspring have no abdomen (posterior)

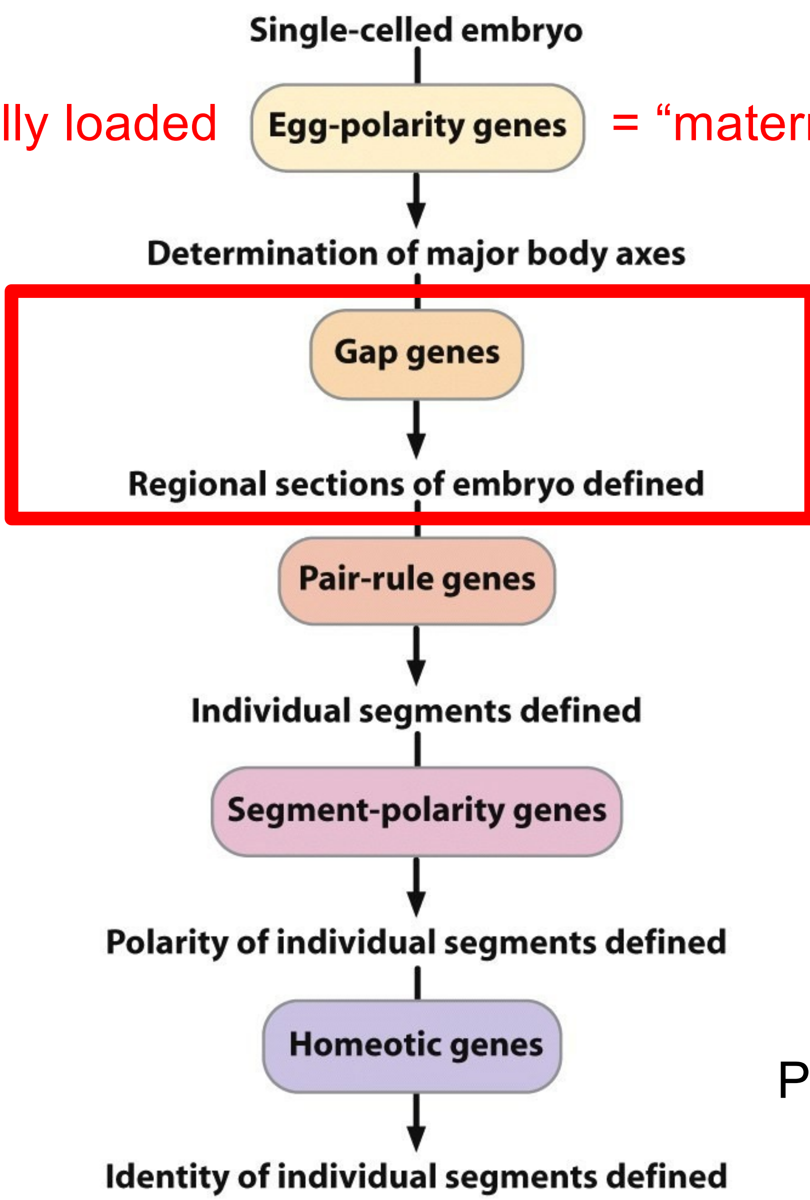
C. 25% of offspring have wild-type appearance

D. 100% of offspring have wild-type appearance

Cascade of transcription factor expression regulates early *Drosophila* development

Maternally loaded

= “maternal effect genes”



Pierce 22.13

Gap genes translate maternal A-P gradients into broad subdomains

- Gap genes (9 identified)
  - e.g. zygotic *hunchback (hb-z), kruppel, knirps, giant*
  - all encode transcription factors
  - gap genes are transcriptionally regulated by maternal-effect gene products

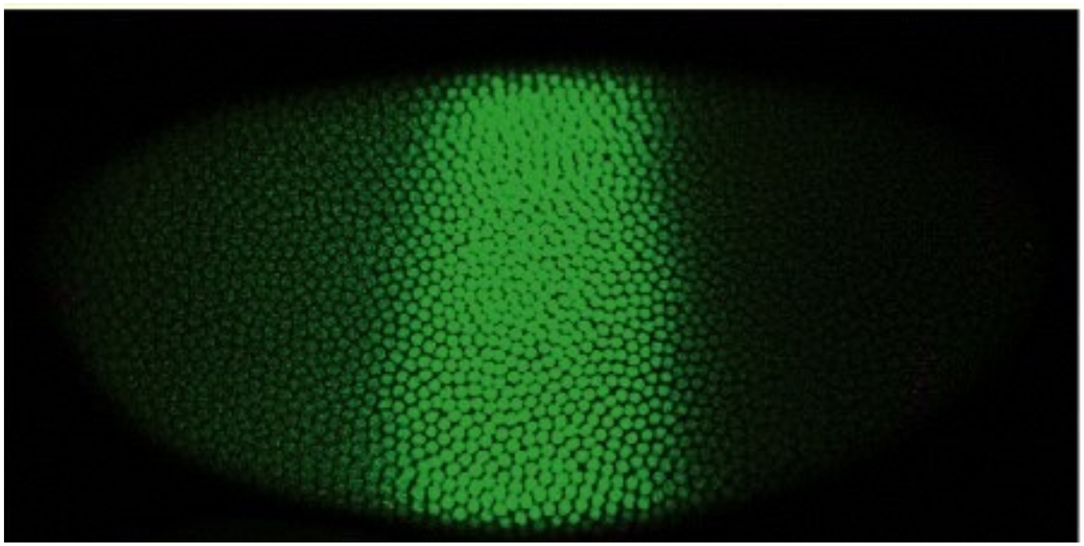


Fig 13-15b – Kruppel protein localization

Gap genes mutants have large gaps in their body plan

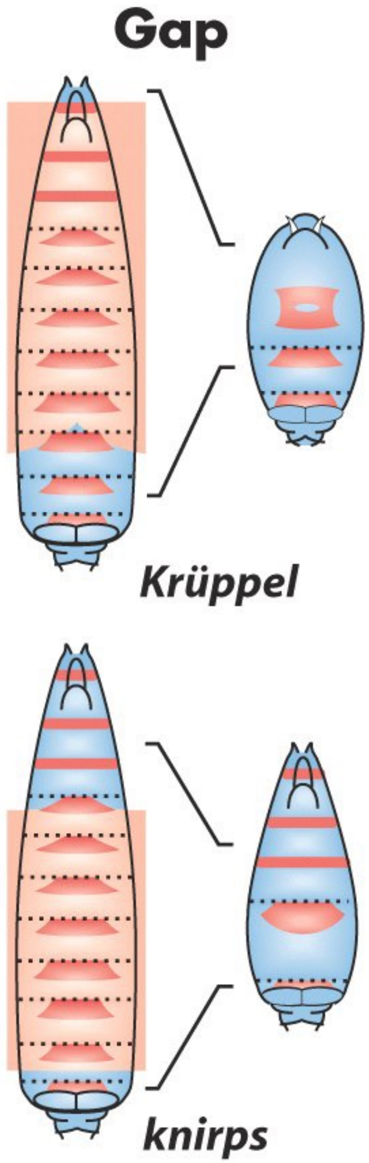
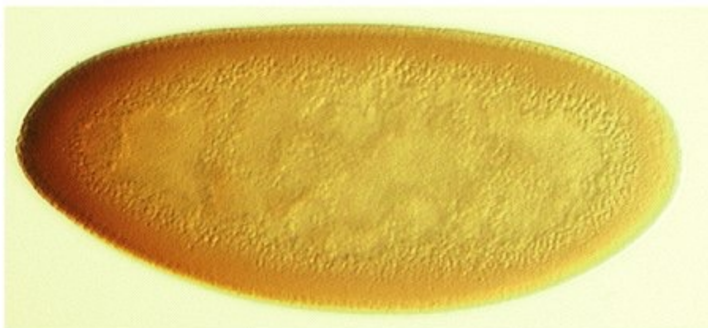


Fig 13-14

gap gene mutants are characterized by loss of several consecutive segments corresponding to the region in the embryo where the gap gene was transcribed

Gap genes are transcriptionally regulated by maternal effect gene products

Bcd protein



Hb-z RNA

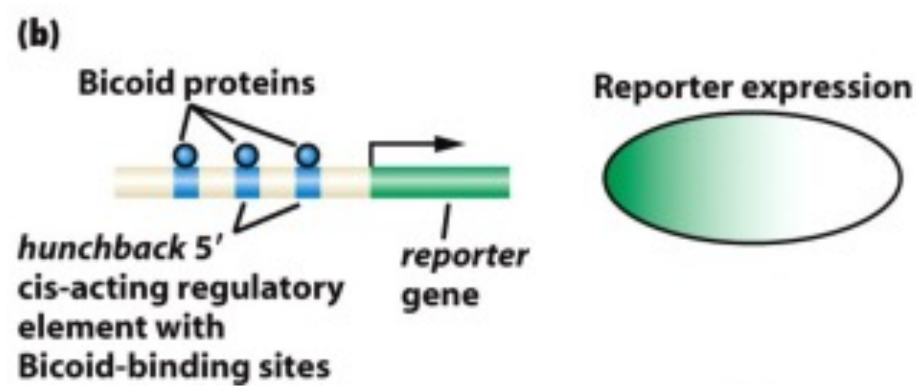
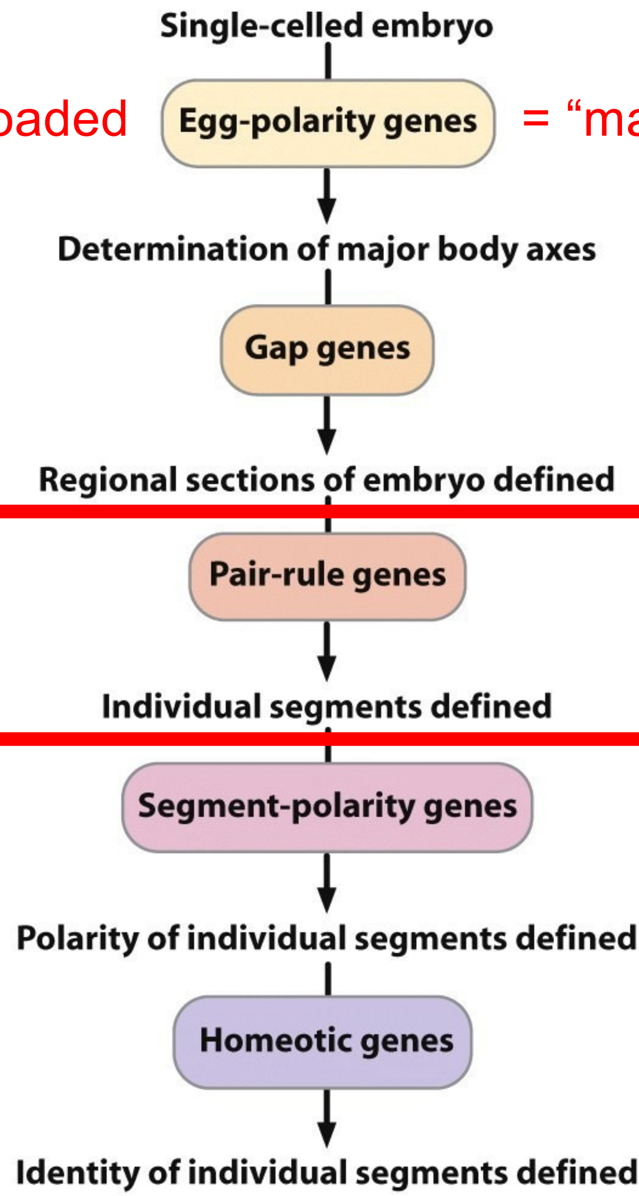


Fig 13-17

- e.g. - zygotic *hunchback (hb-z)*
  - 3 binding sites for Bcd in *hb* promoter

Cascade of transcription factor expression regulates early *Drosophila* development

Maternally loaded



= "maternal effect genes"

Pierce 22.13

Further subdivision of the embryo: Pair-rule genes

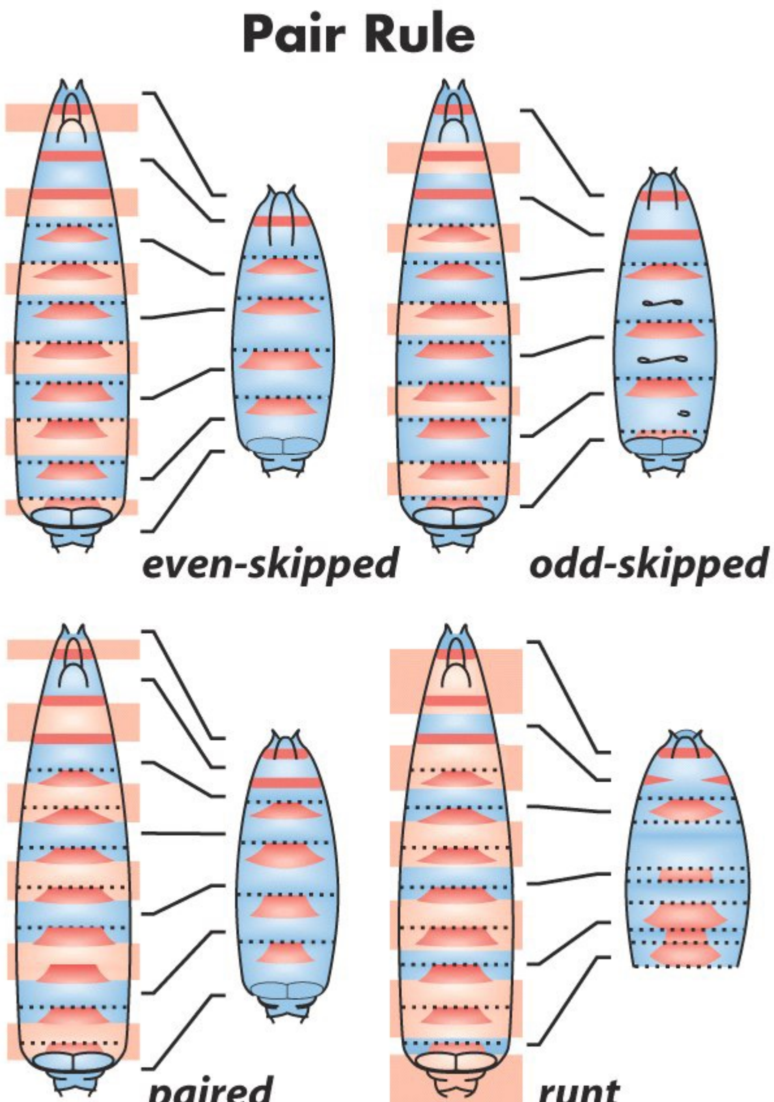


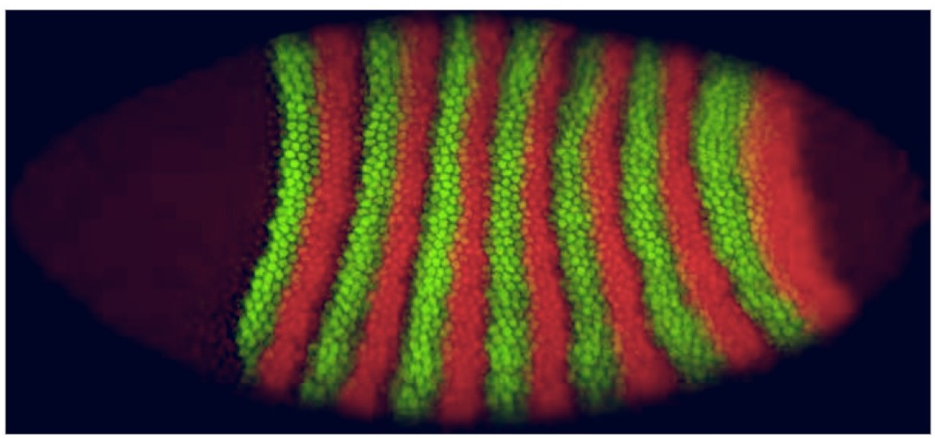
Fig 13-14

- pair-rule gene mutants are characterized by absence of every other segment

Further subdivision of the embryo: Pair-rule genes

- Pair-rule genes (8 identified)
  - all transcription factors
  - each expressed in 7 stripes, although the positions of the stripes are shifted depending on the gene – e.g. Eve vs Ftz:

Overlap of pair-rule stripes



even-skipped fushi tarazu

This arrangement of 7 alternating stripes (pair-rule gene ON/pair rule gene OFF) helps define the 14 segments of the embryo

Pair-rule gene expression is regulated by maternal effect and gap gene

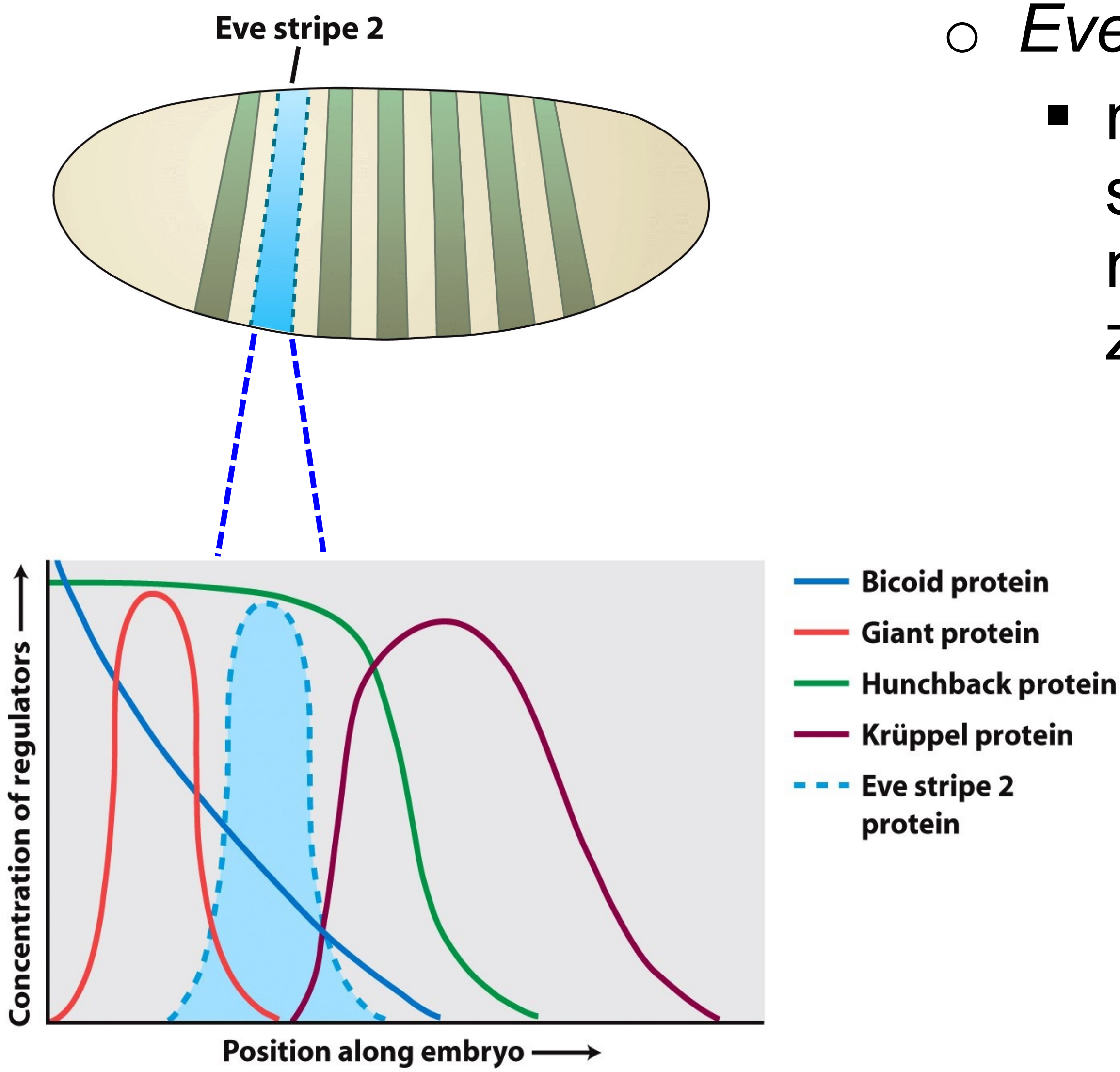


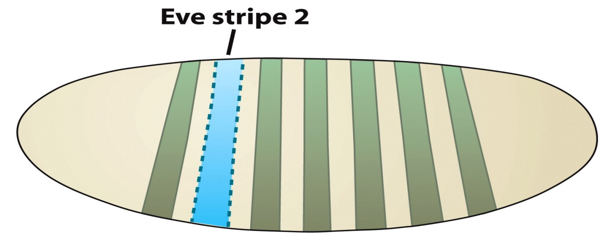
Fig 13-19

- *Even-skipped (eve)*
  - nuclei in stripe 2 will have specific concentrations of maternally loaded and zygotic factors

High [Hb-z]
Medium [Bcd]
Low [Gt], [Kr]

Stripe-specific *eve* enhancers

- *even-skipped (eve)*



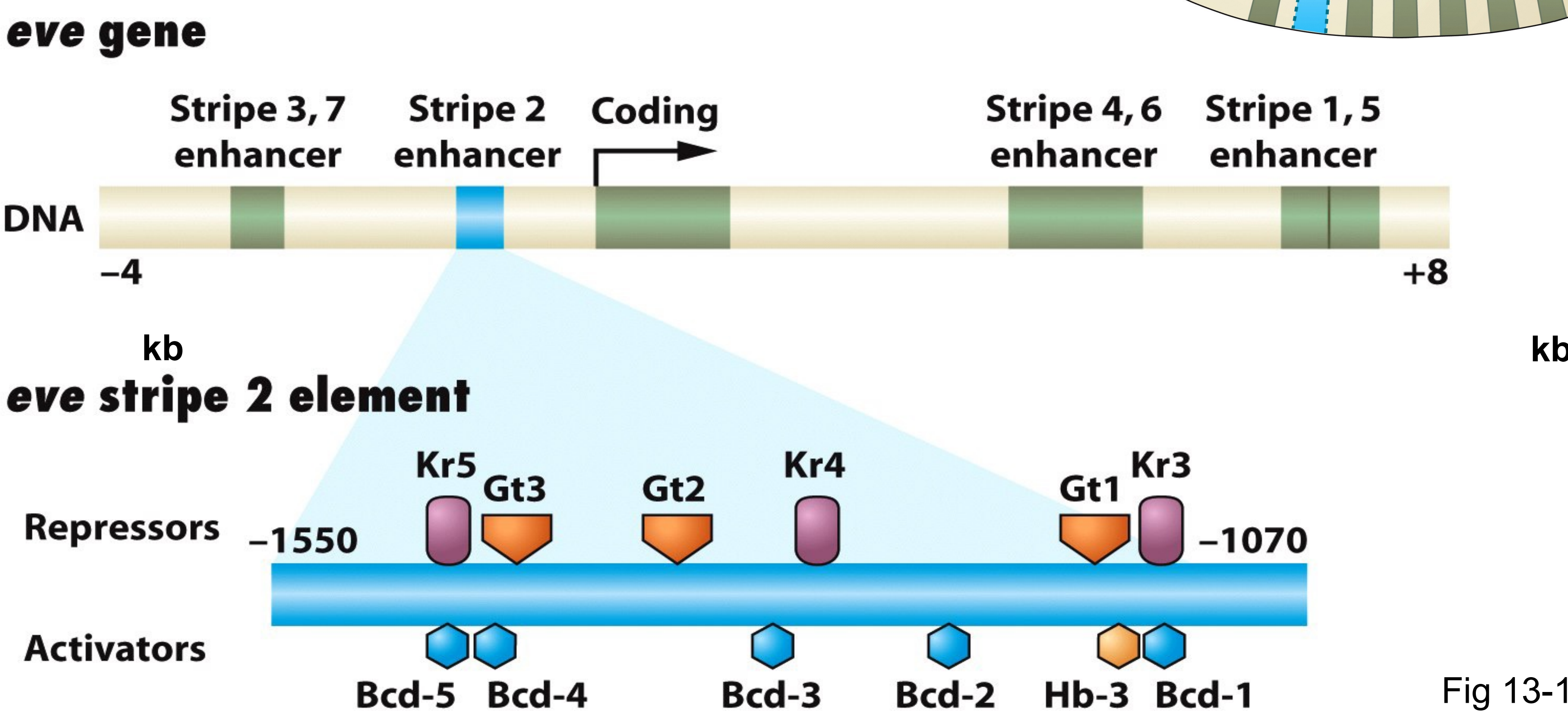
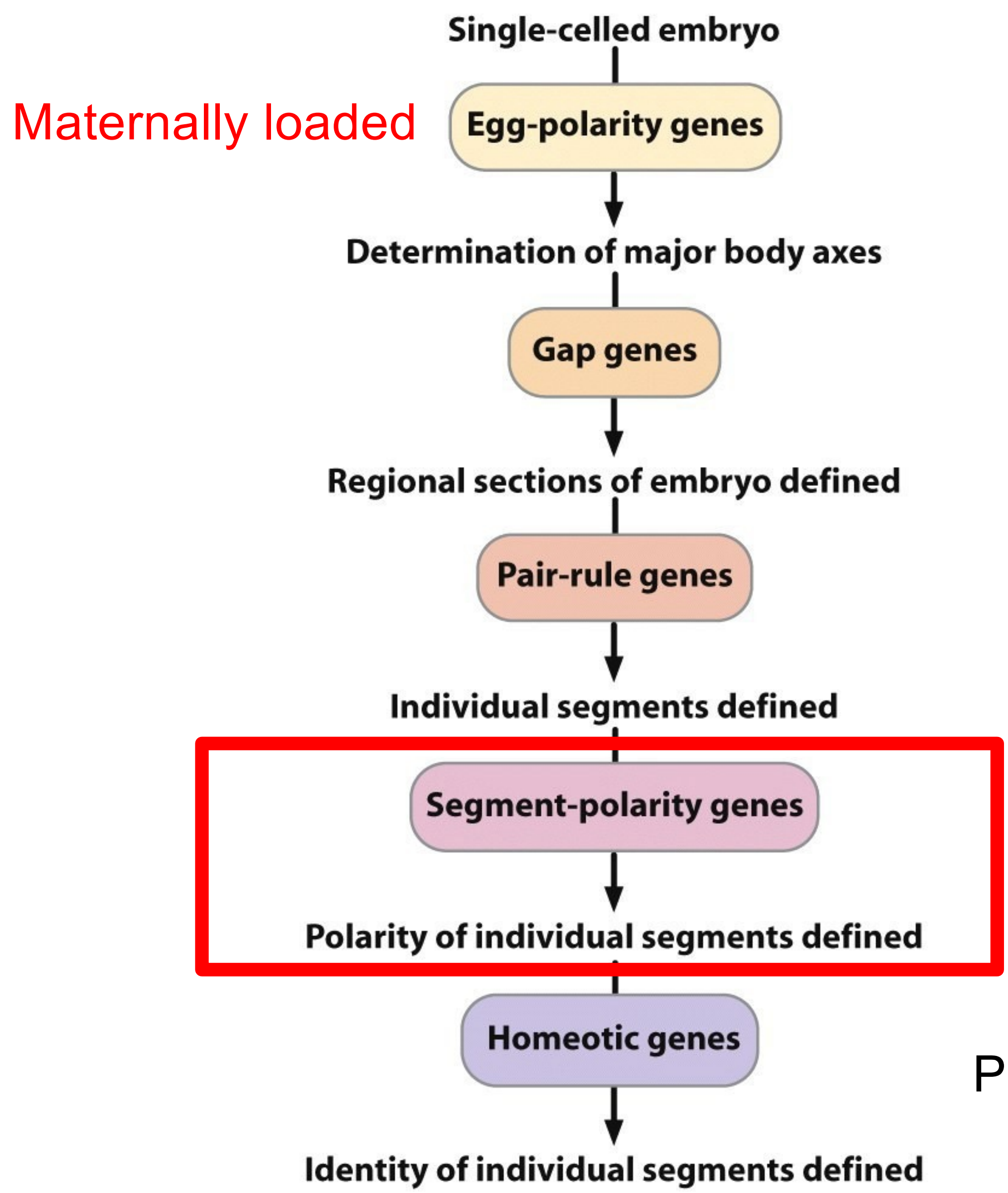


Fig 13-19

Each enhancer has different arrangements of binding sites for maternally loaded factors and Gap genes. This allows enhancers to activate *eve* transcription in a specific stripe of nuclei.

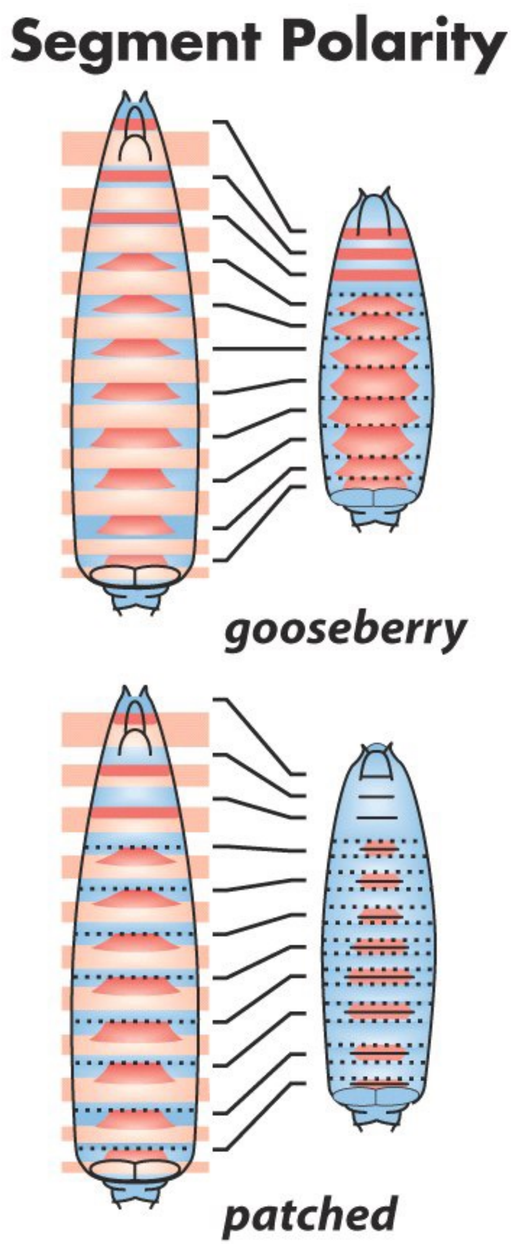
"COMBINATORIAL CONTROL OF TRANSCRIPTION"

Cascade of transcription factor expression regulates early *Drosophila* embryogenesis



Pierce 22.13

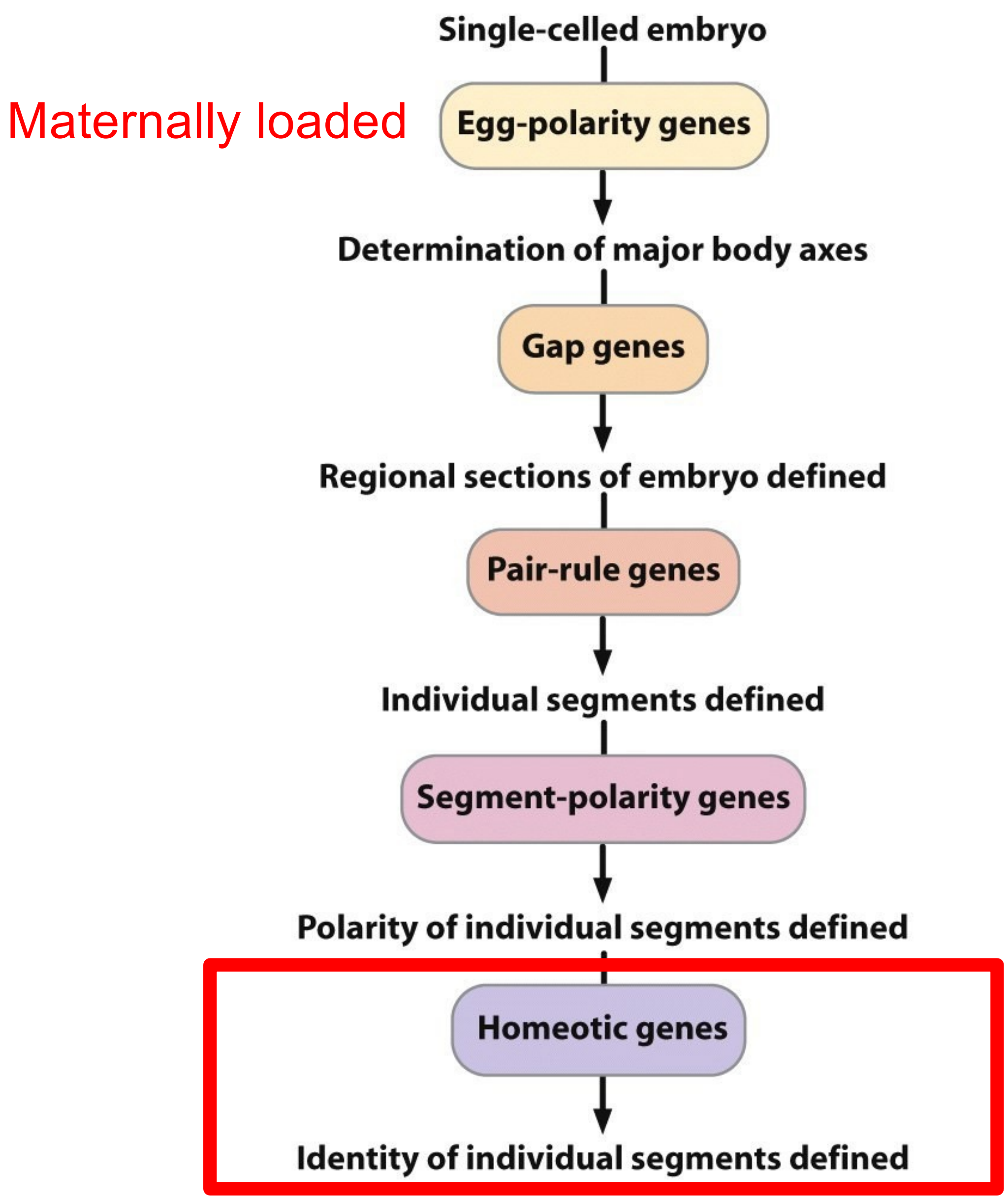
Establishment of segmentation domains



- Segment polarity genes
  - encode components of two cell-cell signaling pathways (Hedgehog, Wingless); includes secreted proteins, membrane receptors, transcription factors…
  - activated/repressed by pair-rule genes
  - Function to define A and P <u>within each segment</u>… mutants have mirroring of one half of each segment

Fig 13-14

Cascade of transcription factor expression regulates early *Drosophila* embryogenesis



Pierce 22.13

Homeotic mutations

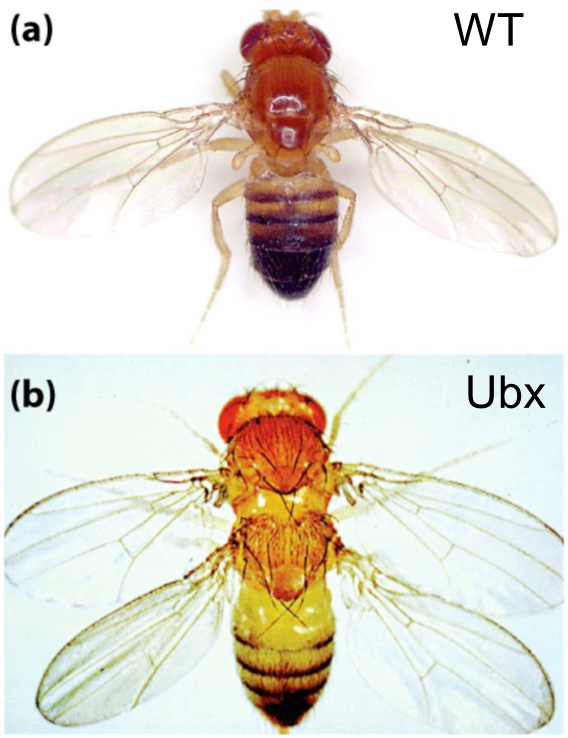
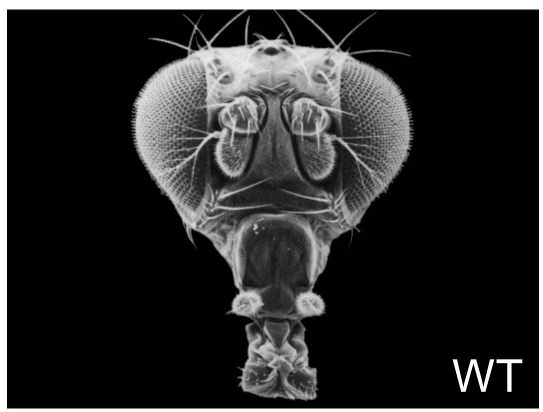
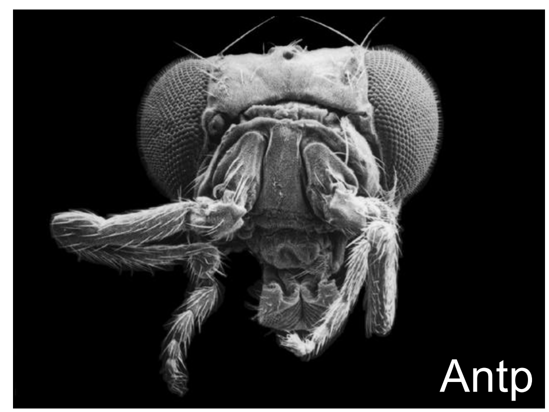


Fig 13-1

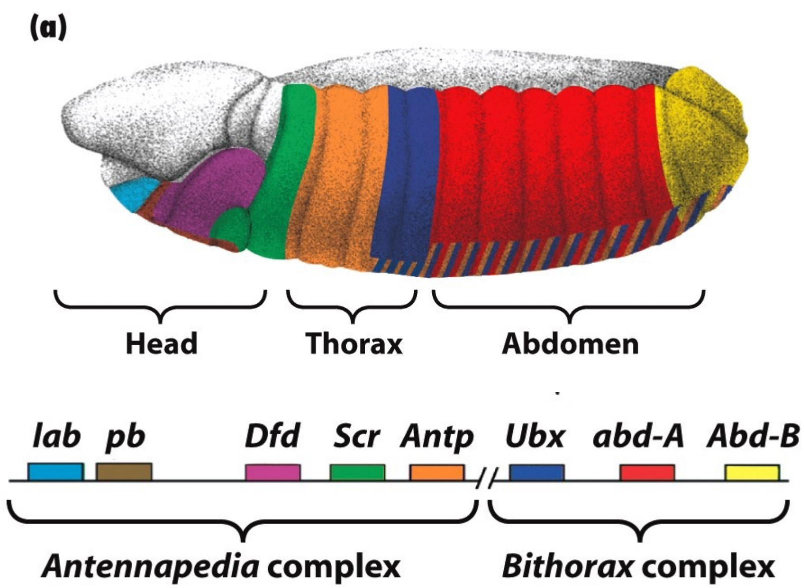
- Mutant animals lack a particular structure, which is replaced by another structure normally found other body segments (homeotic transformation).
- Ultrabithorax (Ubx) – discovered in 1915; second thorax and set of wings in place of halteres
- Antennapedia (Antp) – leg instead of antenna





Establishment of segment identity: *hox* genes

Fig 13-6 - Expression of *hox* genes:



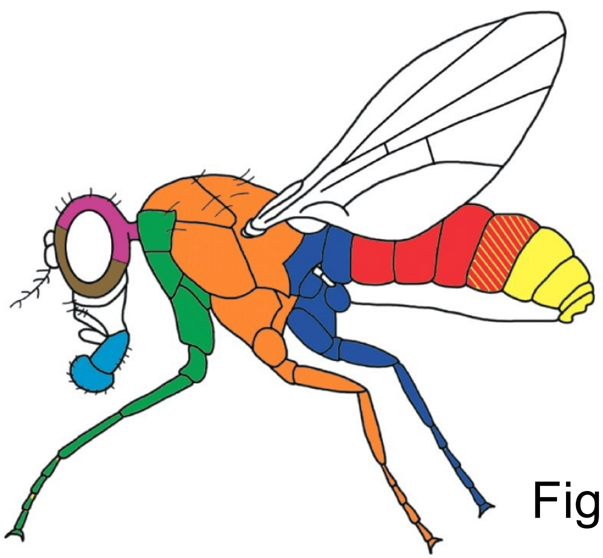


Fig 13-5

- 8 genes, all encode **homeo-domain family transcription factors**
- expressed in specific segments, some overlap
- activated/repressed by gap and pair-rule gene products
- get different structures depending on which *hox* genes are expressed in a segment

The details differ, but the general principles of transcriptional control of development are conserved…

failure of the anterior "organizer" in mice with knockout of the Lim-1 transcription factor

Developmental Biology. 6th edition. Gilbert SF.
Sunderland (MA): Sinauer Associates; 2000.